

PH.D. APPLIED QUALIFYING EXAM 2020 (TAKE HOME)

IMPORTANT - PLEASE READ THESE INSTRUCTIONS CAREFULLY BEFORE YOU START:

This is the Applied Part of the First-Year PhD Qualifying Exam in Statistics. You are permitted to consult textbooks and other scholarly resources to help you in completing the assignment, but you **may not consult or discuss these problems in any way with anyone during the time that the exam is being administered**. If you draw on information from any outside resources (i.e., resources other than class material and notes) then you should clearly cite the source in your write-up.

You should complete the exam alone, in a room separate from any other test-takers. If you have technical questions about the two exam problems, you may email Prof. Sijian Wang at sijian.wang@stat.rutgers.edu (for problem 1) and Prof. Ying Hung at yhung@stat.rutgers.edu (for problem 2), cc'ing Prof. Harry Crane (hcrane@stat.rutgers.edu) AND Prof Tirthankar Dasgupta (td370@stat.rutgers.edu). If you have questions regarding logistics (e.g., submission instructions given below), please email Prof. Harry Crane (hcrane@stat.rutgers.edu) AND Prof Tirthankar Dasgupta (td370@stat.rutgers.edu).

Submission Instructions:

- (1) Type up your answers to the two problems in two separate Word or Latex files.
- (2) Create a folder for each problem. The folder must contain the following:
 - (a) A word or pdf file containing your answers to the problem, including all tables, figures and Appendices.
 - (b) An R (or python) file containing the codes used to generate your results.
 - (c) A word or a text file that gives a summary of the codes submitted.
- (3) Place the two folders into a single folder titled **yourlastname_yourfirstname_Appliedexam2020**. Zip this folder and email it to hcrane@stat.rutgers.edu AND td370@stat.rutgers.edu by 5 PM July 24, Friday.

Problem 1

BACKGROUND

Alzheimer's disease (AD) neuropathology, including beta-amyloid plaques, neurodegeneration and neurofibrillary tangles, begins to develop at least 10-20 years before a clinical diagnosis of dementia is given. One hypothesis regarding the reason for failed clinical trials is that the treatment is initiated too late in the disease, after neuronal death has occurred. With novel technologies, it may now be possible to identify markers (predictors) of AD during a preclinical disease phase prior to symptom onset. For example, cerebrospinal fluid (CSF) acquired from the lumbar section of the spinal cord can be assayed for proteins associated with beta-amyloid plaques (ab_{1.42}), neurodegeneration (total tau [ttau]) and neurofibrillary tangles (phosphorylated tau [ptau]), all of which are in turn associated with AD.

In individuals with AD, the typical CSF profile is low ab_{1.42} (reflecting amyloid protein accumulated in plaques in the brain rather than releasing through the CSF) and high total and phosphorylated tau (reflecting breakdown of neurons due to neurofibrillary tangles). An important goal of current studies is to identify AD accurately and reliably in the earliest stages of disease. This enables future clinical trials to enroll participants at highest risk for developing AD to test if early interventions slow or prevent progression of cognitive decline, leading to AD. To do that, it is first necessary to know if biomarkers of disease in preclinical AD accurately predict cognitive decline.

METHODS

Data for the current study are drawn from a dataset collected between 2001 and 2016 in a community sample of about 1500 late middle-aged adults (mean age at baseline = 54). The cohorts were designed to be enriched for higher AD risk (e.g., ~75% with a family history of AD). Participants return for study visits every 2 years (with the exceptions of 4 years between visits 1 and 2 for a subset and 1 year between visits for older participants/participants with AD or mild cognitive impairment (MCI)) and undergo neuropsychological cognitive evaluations and medical exams. Lumbar punctures to obtain CSF were conducted on approximately 500 participants as part of a linked study.

Cognitive status at each study visit was determined by a consensus panel based on review of cognitive performance at all visits and review of additional information in each participant's chart. Possible cognitive status outcomes include: 0) cognitively normal, 1) early MCI, 2) clinical MCI, 3) dementia, or 4) impairment-not MCI. CSF samples were assayed for several proteins, including ab_{1.42}, ttau, and ptau. With the exception of Category 4), these outcomes are ordered: 0), 1),

2) and 3). Patients with Category 4) are not of direct interest and might be deleted for initial analyses.

Note: Mild Cognitive Impairment (MCI) is a state of cognitive impairment that often precedes fulminant dementia. Many participants with MCI progress to AD, however, some individuals stabilize, whereas others even show improvement in cognitive performance back to their personal baseline.

QUESTIONS

Please answer the following questions by clearly describing conclusions (answering the questions below), clearly explaining and justifying statistical models/methods used to answer the questions, interpretation of results (evidence to your answers) from data analysis and how these results lead to your answers. If you are unable to answer questions, recommend what additional information is required to find answers.

- 1) Do CSF biomarkers predict progression from cognitively normal at baseline to cognitively impaired (e.g., early MCI, clinical MCI, or dementia) at most recent visit? Do individual biomarkers or a combination of biomarkers (ratios provided) better predict cognitive status?
- 2) For all participants, what is the course of cognitive status over time as measured by diagnostic category at each visit? How many people worsen (e.g., cognitively normal to MCI or dementia, MCI to dementia), how many improve (e.g., dementia to MCI, MCI to cognitively normal), and how many remain the same? Do CSF biomarker values predict the course of cognitive status during the study?
- 3) Does the relationship between CSF biomarkers and cognitive status vary depending on age, sex, genetic risk for AD (APOE status), parental history of AD, or years of education?
- 4) Does deleting patients who start with or develop “impairment-not MCI” (outcome 4) cause any scientific risks to your answers to the above questions?
- 5) Write a one page summary describing the key scientific questions you addressed in parts 1)-4) above, your approach and overall conclusions.

DATA FILE

The data file AD_dataset_applied2020.csv contains data on the following variables:

`export_id`: De-identified participant ID

`age_at_LP`: age at lumbar puncture

gender: gender

TotalYearsEducation: years of education completed

apoe4pos: APOE E4 status (0=no genetic risk; 1=genetic risk)

Parent_Dementia: 0=no parent with dementia; 1=parent with dementia

LP1_NPv1_absmonthsdiff: Months between lumbar puncture and study visit 1

LP1_NPv2_absmonthsdiff: Months between lumbar puncture and study visit 2

LP1_NPv3_absmonthsdiff: Months between lumbar puncture and study visit 3

LP1_NPv4_absmonthsdiff: Months between lumbar puncture and study visit 4

LP1_NPv5_absmonthsdiff: Months between lumbar puncture and study visit 5

LP1_NPv6_absmonthsdiff: Months between lumbar puncture and study visit 6

Diagnosis_v1: Cognitive status at visit 1 (0=Cognitively Normal; 1=Early MCI; 2=Clinical MCI; 3=Dementia; 4=Impaired not MCI)

Diagnosis_v2: Cognitive status at visit 2 (0=Cognitively Normal; 1=Early MCI; 2=Clinical MCI; 3=Dementia; 4=Impaired not MCI)

Diagnosis_v3: Cognitive status at visit 3 (0=Cognitively Normal; 1=Early MCI; 2=Clinical MCI; 3=Dementia; 4=Impaired not MCI)

Diagnosis_v4: Cognitive status at visit 4 (0=Cognitively Normal; 1=Early MCI; 2=Clinical MCI; 3=Dementia; 4=Impaired not MCI)

Diagnosis_v5: Cognitive status at visit 5 (0=Cognitively Normal; 1=Early MCI; 2=Clinical MCI; 3=Dementia; 4=Impaired not MCI)

Diagnosis_v6: Cognitive status at visit 6 (0=Cognitively Normal; 1=Early MCI; 2=Clinical MCI; 3=Dementia; 4=Impaired not MCI)

Diagnosis_v7: Cognitive status at visit 7 (0=Cognitively Normal; 1=Early MCI; 2=Clinical MCI; 3=Dementia; 4=Impaired not MCI)

LP1_Diagnosis: Cognitive status at visit closest to lumbar puncture (0=Cognitively Normal;

1=Early MCI; 2=Clinical MCI; 3=Dementia; 4=Impaired not MCI)

pv_ln_ab_1_42: ab_1_42 (log-transformed)

pv_htau: ttau (total tau)

pv_ptau: ptau (phosphorylated tau)

pv_ptau_by_ab: ptau_by_ab (ratio of phosphorylated tau to ab_1_42)

pv_ttau_by_ab: ttau_by_ab (ratio of total tau to ab_1_42)

pv_ab42_by_ab40: ab42_by_ab40 (ratio of ab_1_42 to ab_1_4 ; a normalized ab value)

Notes:

- Impaired not MCI = impaired performance on neuropsychological tests, but due to other factors (e.g., fatigue, poor effort, lifelong learning disability). Do not include in analyses.
- 999= missing data

Problem 2

- Problem 2.1 (a) Let (X, Y, Z) be a three-dimensional random vector following probability density function $f(x, y, z)$, assumed to be known. Describe Gibbs sampling algorithm to simulate/generate realizations of the random vector (X, Y, Z) .
- (b) Show that Gibbs sampling is a special case of Metropolis-Hastings algorithm.

Problem 2.2 Piston slap is an unwanted engine noise that results from piston secondary motion and is one of the leading causes of dissatisfaction of automobile customers. In an attempt to predict piston slap noise, an engineer considers six input variables: clearance between the piston and the cylinder liner (x_1), location of peak pressure (x_2), skirt length (x_3), skirt profile (x_4), skirt ovality (x_5), and pin offset (x_6). Two sets of data are provided in text file format – one for training (piston slap noise training data.txt) and the other for testing (piston slap noise test data.txt). Perform the following analysis:

- (1) Fit a smoothing spline for the piston slap noise as a function of x_6 with the smoothing parameters specified by cross-validation. Report the estimated smoothing parameters and plot the fitted model as a function of x_6 .
- (2) Use a bootstrap method to construct a bootstrapped confidence band for the smoothing spline obtained in (1).
- (3) Consider a Gaussian process model:

$$Y(\mathbf{x}) = \mathbf{x}^T \boldsymbol{\beta} + Z(\mathbf{x}),$$

where $\mathbf{x} = (x_1, \dots, x_6)^T$, $\boldsymbol{\beta}$ is a vector of unknown coefficients, $Z(\mathbf{x})$ is a Gaussian process with mean 0 and covariance function defined by $cov(Y(\mathbf{x}), Y(\mathbf{x} + \mathbf{h})) = \sigma^2 \phi(\mathbf{h}; \boldsymbol{\theta})$, and $\boldsymbol{\theta}$ is a vector of unknown correlation parameters. Define the correlation function $\phi(\mathbf{h}; \boldsymbol{\theta})$ by

$$\phi(\mathbf{h}; \boldsymbol{\theta}) = \exp\left(-\sum_{i=1}^6 \theta_i h_i^2\right).$$

Write down the explicit form of the MLE of parameters $\boldsymbol{\beta}$, $\boldsymbol{\theta}$, and σ^2 and compute the estimated values, writing your own R code (DO NOT use any standard R package for fitting GP regression like mlegp or dicekriging).

- (4) Construct the best linear unbiased predictor (BLUP) based on (3) and predict the piston slap noise corresponding to each data point in the testing set. Summarize the prediction performance using suitable performance measures and comment.